

Vibration-Based Condition Monitoring and Fault Diagnosis

Vibration-based Condition Monitoring (VCM)

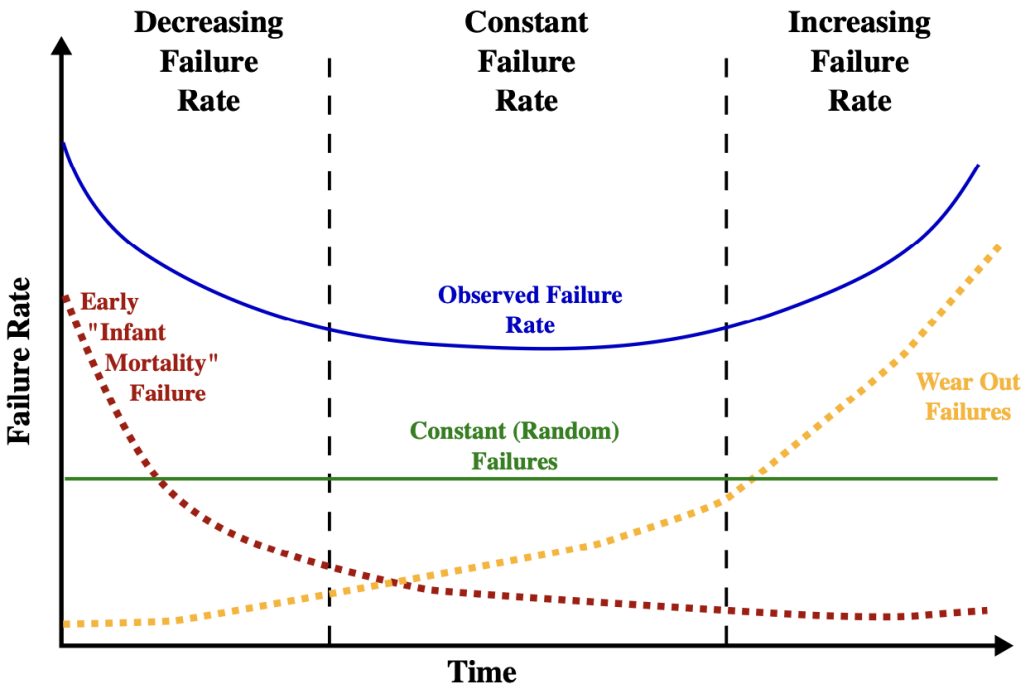
A tool to identify the fault(s) so that the identified fault can be fixed during the planned shutdown.

Investigating the reason for the frequent and repetitive failures in any machines by doing:

- Additional vibration tests
- Analysis of the measured vibration data
- Regular machine condition monitoring.

This lecture provides the basic concept on the industrial vibration-based condition monitoring (VCM) for any rotating machines.

DIFFERENT STAGES OF VIBRATION MEASUREMENTS AND MONITORING



Life Cycle of Each Machine:

Early failure, Useful life, and Old age.

Time span of each stage can be different even for the identical machines and structures.

CM should be applied during the **complete life cycle** for critical machines.

For simplification, the VCM approach for any machines is divided into three stages

- stage 1: Machine installation and commissioning
- stage 2: Machine Operation
- stage 3: Aged machines



STAGE 1-MACHINE INSTALLATION AND COMMISSIONING

Early failure **may not be related** to their poor design in most cases.

Identical machines may be working perfectly fine within the same plant or in many other industries. Machine generally consists of

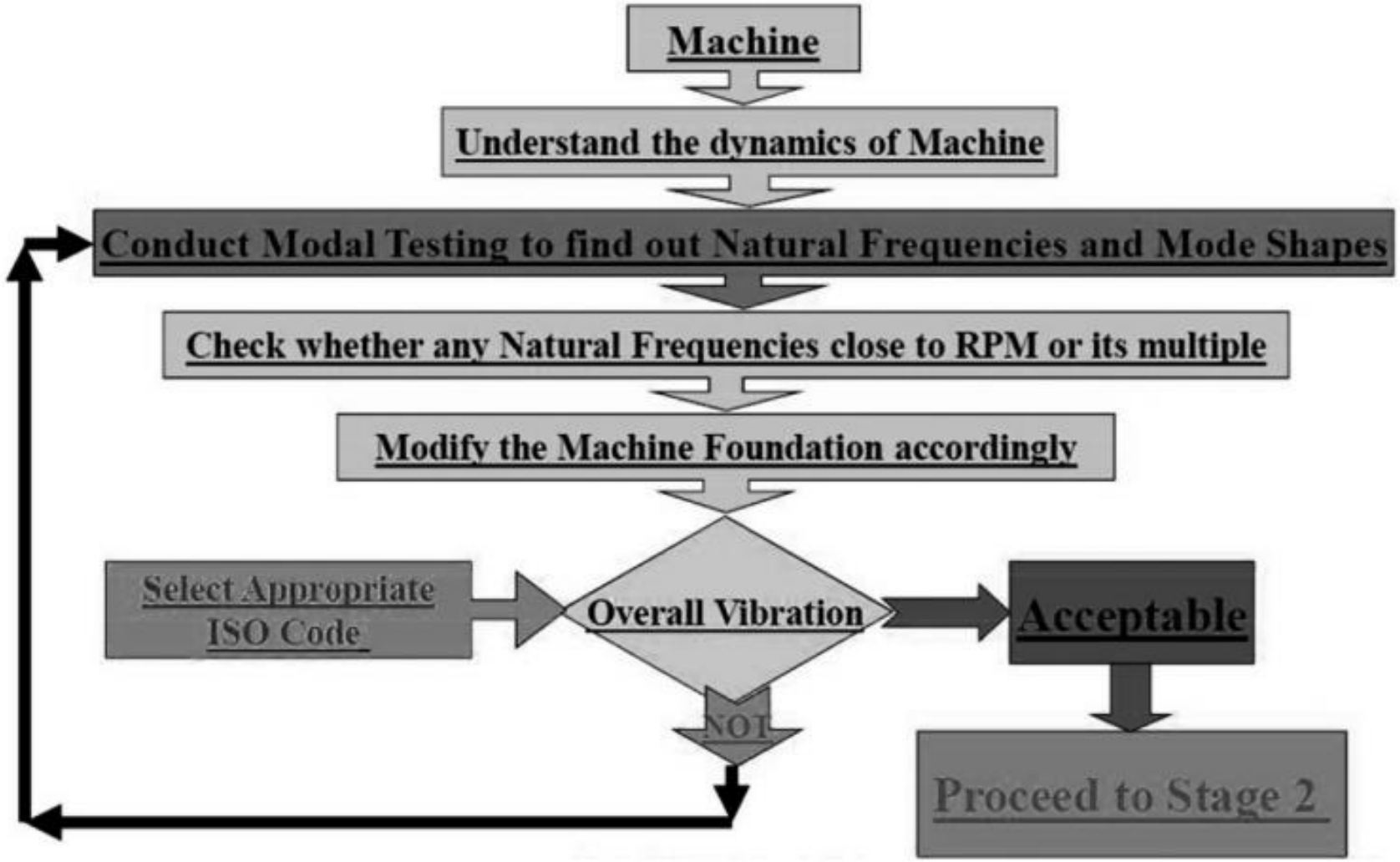
1. rotor,
2. bearing
3. foundations including all auxiliary components.

- The assembly and foundation of any machine during installation generally plays significant role in the early failure
- foundation may not be identical either due to space limitation or due to easy and convenient approach used during installation

STAGE 1-MACHINE INSTALLATION AND COMMISSIONING

- Dynamics of any machine is the combinedd effect of the rotor, bearing and foundation.
- One of the machine natural frequencies of the as-installed machine is close to machine RPM or its multiple harmonics, results into the resonance during machine normal operation and leads to frequent failures due to high vibration problem.
- The high vibration problem during installation and commissioning should not be solved using trail and error approach. It is better to understand the dynamic of the Machine.

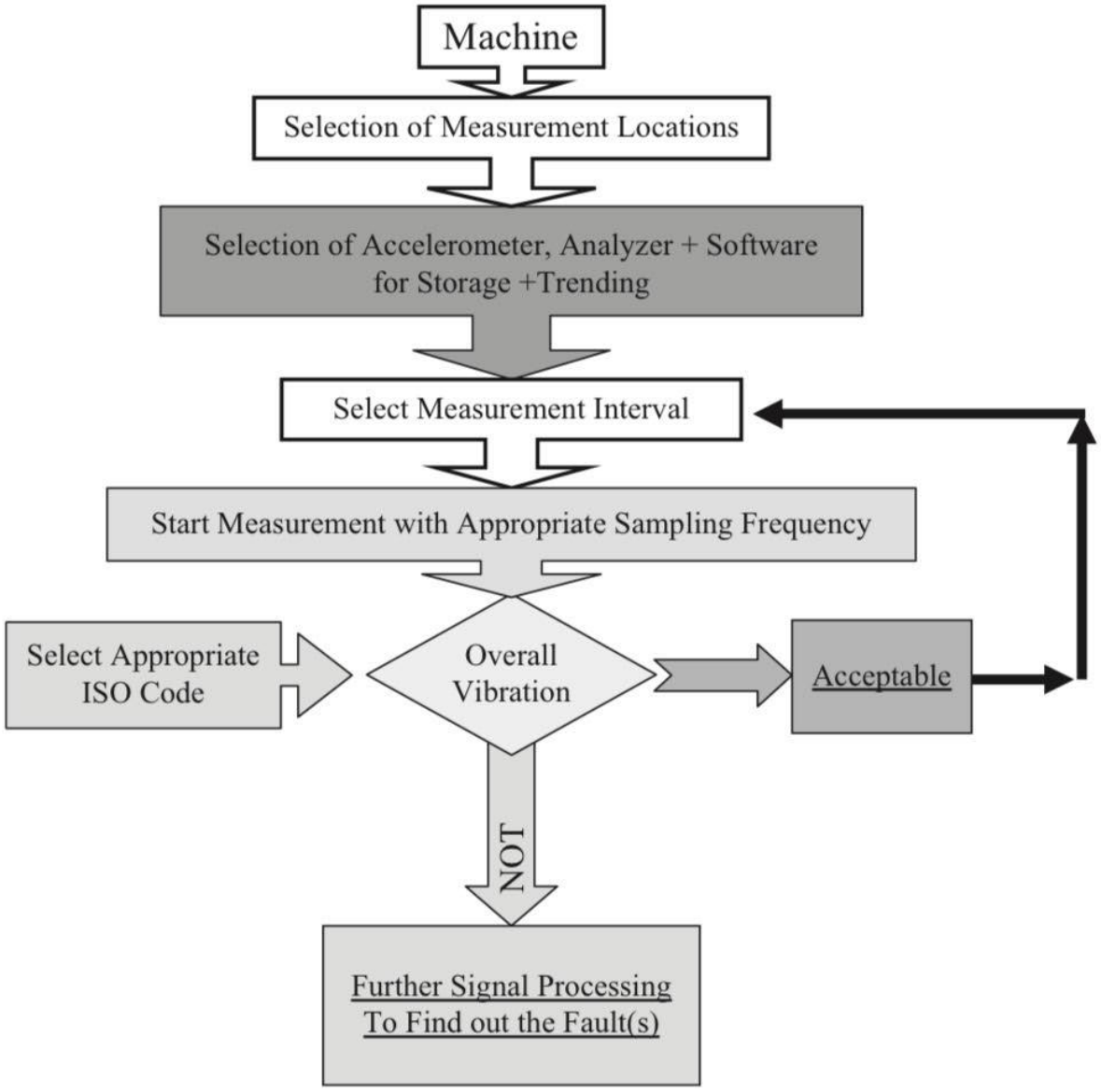



Machine
Understand the dynamics of Machine
Conduct Modal Testing to find out Natural Frequencies and Mode Shapes
Check whether any Natural Frequencies close to RPM or its multiple
Modify the Machine Foundation accordingly
Select Appropriate ISO Code
Overall Vibration
NOT
Acceptable
Proceed to Stage 2

If the vibration of the machine is within acceptable range during installation and commissioning:

1. Apply regular VCM
2. Few defects may randomly develop during regular operation
3. Early identification is important to rectify during planned shut down.

Stage 2: Machine Operation

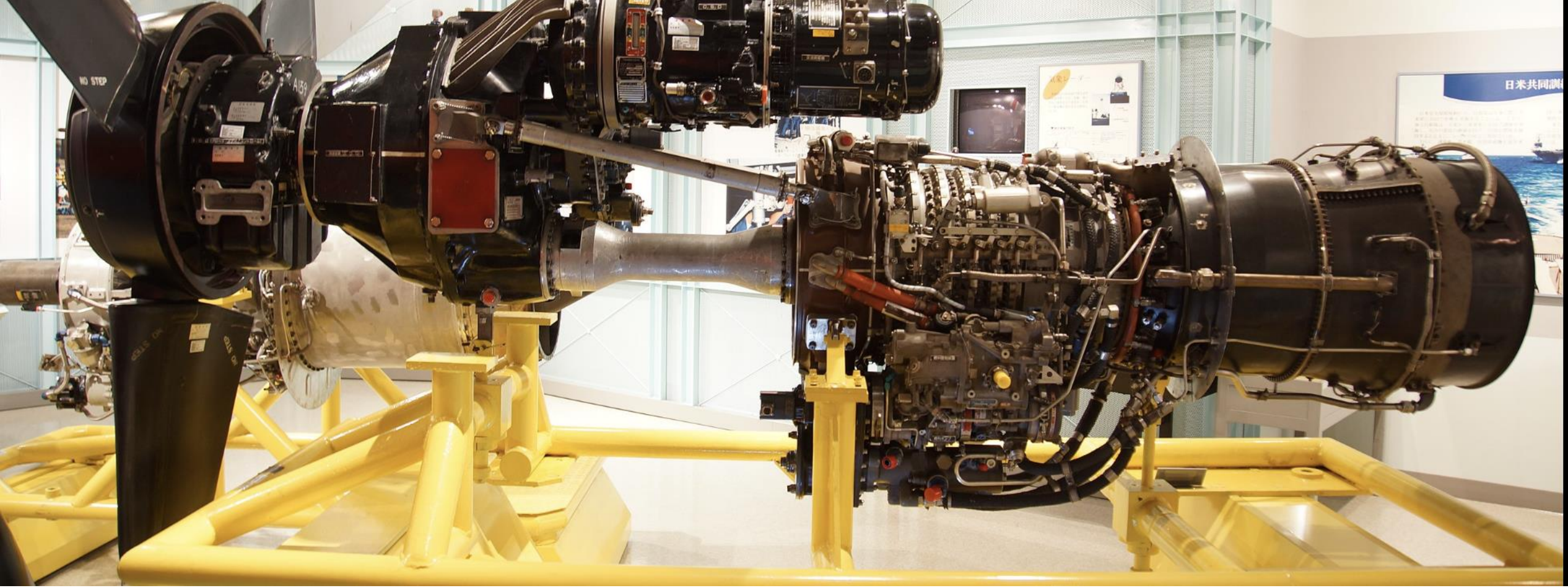


Machine Characteristics for VCM

Before planning carrying out vibration measurements, following info is required:

- Power rating so that appropriate ISO code can be selected for the vibration severity limit. ISO 10816 and 20816 codes may be useful for this purpose.
- Number of bearings and their types. For anti-friction bearings full details must be known to estimate the defect-relate bearing characteristics frequencies
- Clearance, type, and other details in case of fluid bearings.





Machine Characteristics for VCM

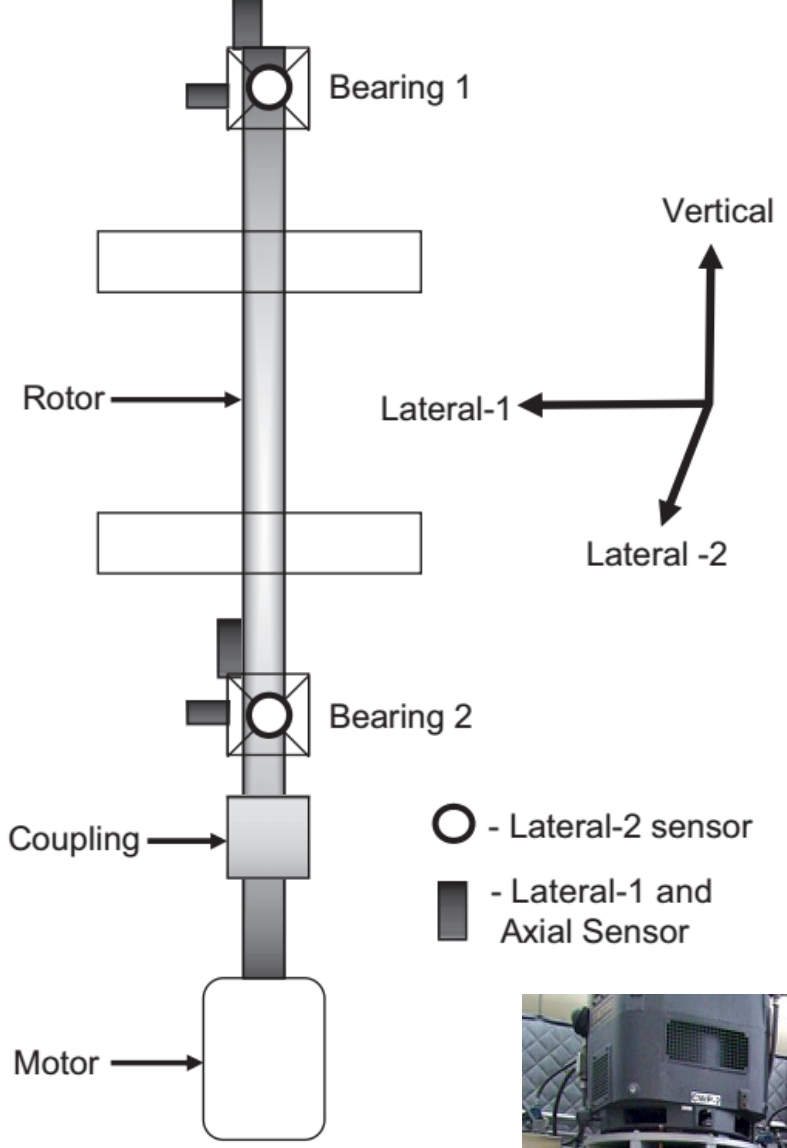
I. Number of impeller blades and number of stages
II. Details of gear box
III. Machine RPM
IV. Motor specifications if it is driven by a motor

Measurement Location for VCM

Bearing housing/ pedestal in three directions is preferred location for the measurement as the measured vibration responses are expected to have responses from the rotor, bearing, gearbox, etc.

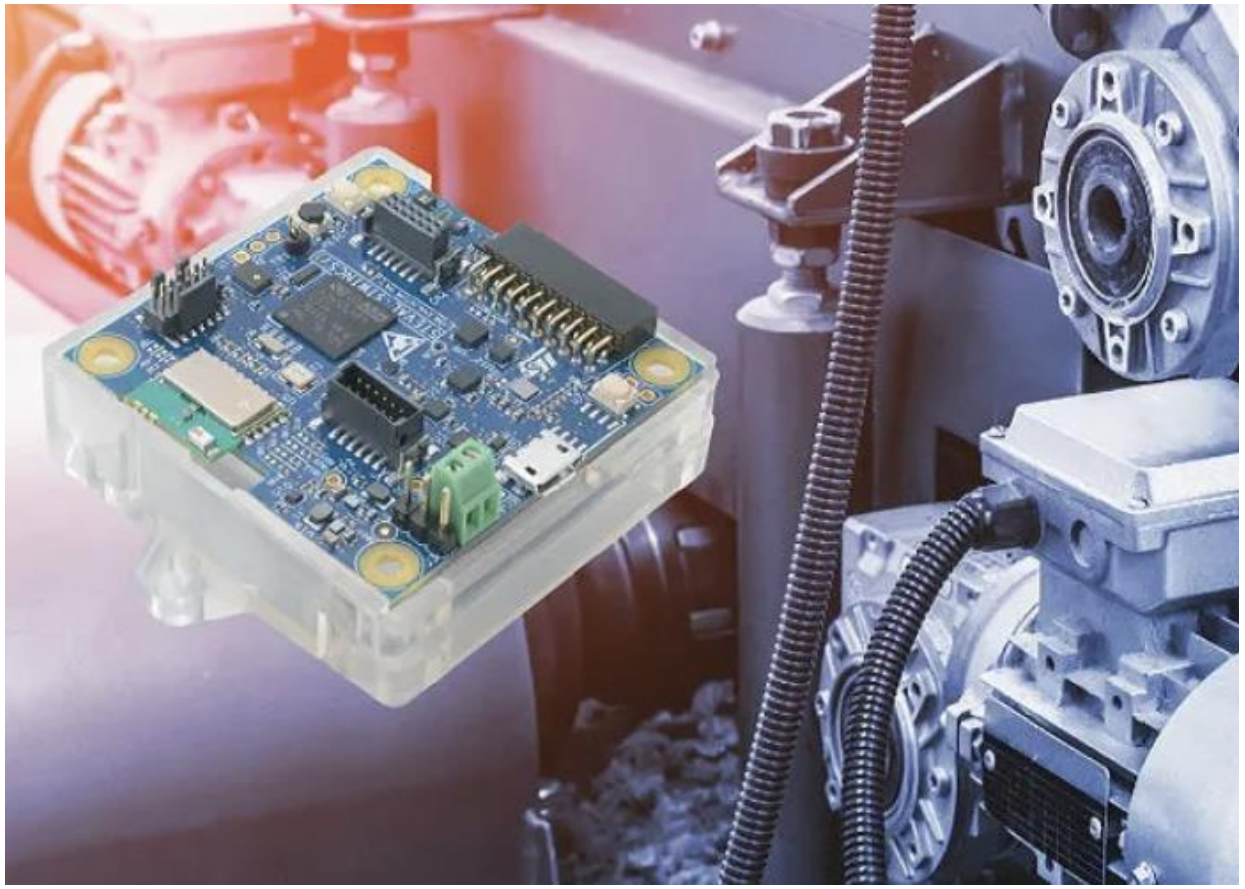
measurements in three orthogonal directions will map the dynamic behavior of any machine,

1. For horizontal machines, the measurement should be taken in the axial, vertical and lateral directions.
2. For vertical machines, the measurements should be in the axial and two-lateral directions as shown.





Software and Measurement Interval in VCM



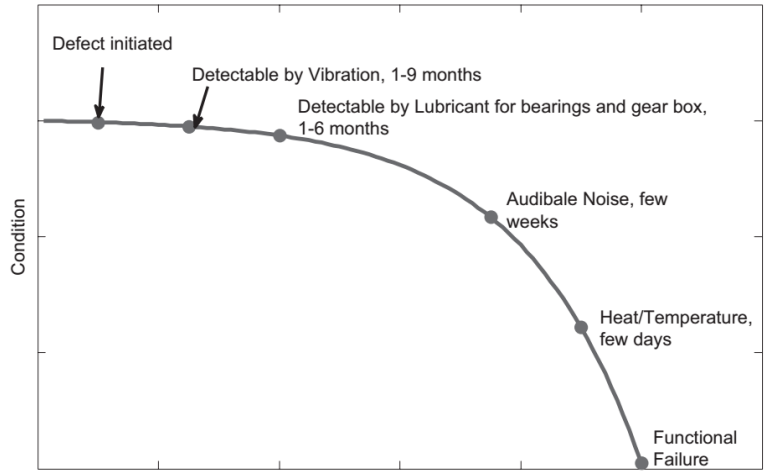
Software and instrumentations:

Appropriate selection of instrumentation such as sensors, data collections and their analysis either through hardware or software codes are essential for the meaningful VCM.

Measurement interval:

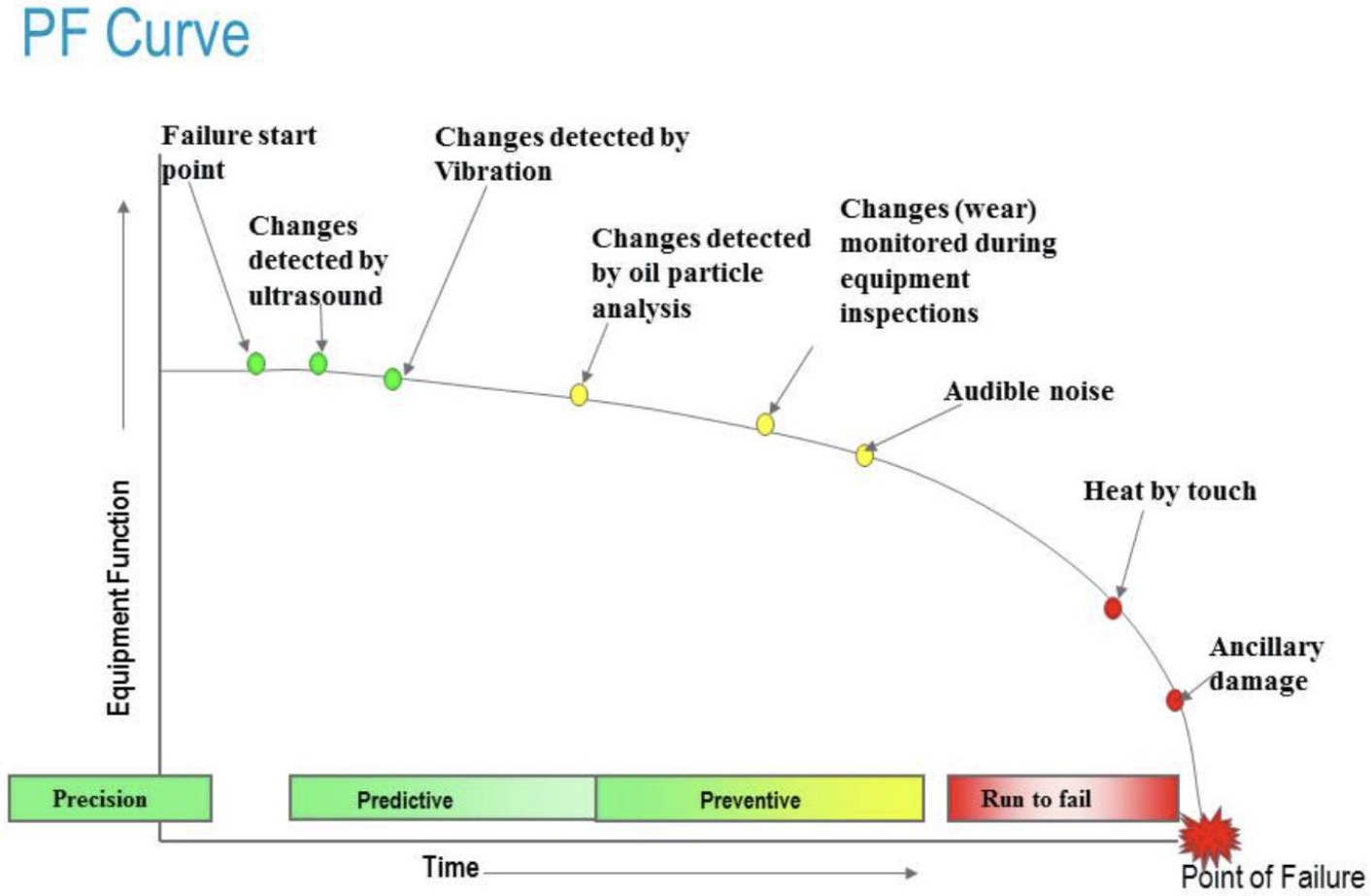
to decide whether the measurement is required continuously using online system once per day or at the interval of few days.

Measurement Interval in VCM



The P-F (Potential Failure) curve is well-known concept that represent the behavior of asset conditions from the "initiation of defect" to the "failure.

1. VCM has capability to identify fault at much early stage and provide the lead time to failure ranging from month and more.
2. Machines are not going to fail immediately even if the fault has already initiated. It needs to go through number of vibration cycles before leading to failure.
3. If machine shows increase in trend it is better to do more frequent measurement.





VIBRATION SEVERITY PER ISO 10816						
Machine			Class I small machines	Class II medium machines	Class III large rigid foundation	Class IV large soft foundation
	in/s	mm/s				
Vibration Velocity Vrms	0.01	0.28				
	0.02	0.45				
	0.03	0.71		good		
	0.04	1.12				
	0.07	1.80				
	0.11	2.80		satisfactory		
	0.18	4.50				
	0.28	7.10		unsatisfactory		
	0.44	11.2				
	0.70	18.0				
	0.71	28.0		unacceptable		
	1.10	45.0				

Vibration Severity limits

Overall vibration limit in RMS velocity (Vrms) of machine vibration

1. ISO10816
2. ISO20816

- The use of correct frequency band in the vibration measurement is very important to cover all fault frequencies so RMS value reflect machine condition.
- It is the machine manufacturer's responsibility to provide the vibration limits to stablish the allowable and alarm limits



Vibration Measurement Guideline

a pump driven by a 2kW motor is up to 6 kHz:

- Vibration velocity, V_{RMS} is 1 mm/s (above allowable limit of 0.71 mm/s) when measured up to 6 kHz. ->needs further investigation.
- vibration velocity, VRMS is 0.5 mm/s (well below allowable limit of 0.71 mm/s) when measured up to 2 kHz. -> safe to operate

choose the frequency range for any machines carefully prior to the vibration measurements.

Stage 3: Aged Machines

- Many identical machines continue to function properly after many years of operations
- some starts failing frequently e.g., frequent failure of bearings, gearbox, etc. After 15-20 years of operation.

It indicates that some structural changes may have happened over the period of operation which has resulted in frequent and repeated failures.

